Release Note SW

Product designation:	MG1CS022_62XSC / V036_TPROT
Version:	1.0
Date:	2026-03-25
Customer:	HONDA

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1 Product identification

This Release Note SW describes the SW which can be flashed in the below described HW.

Product designation:	MG1CS022_62XSC / V036_TPROT
Product part number:	10SW132477 10SW132478 10SW132479 10SW132480 10SW132481 10SW132482
Project:	Honda_AP2_MY27_City_MG1
Name of customer:	HONDA
Supported Hardware name/Part number:	MG1CS022-3.30 / 0261B00CE7, 0261B00CF1, 0261B00CE8, 0261B00CE9, 0261B00CF2, 0261B00CF0
Program version reference in the Bosch Configuration Management system:	PVER : MG1CS022_62XSC / V036_TPROT
Program version maturity:	Series
Release status Software:	Released for series production
Usage/Remarks	This software is released for series production
Predecessor program version (development baseline):	PVER : MG1CS022_62XSC / V036
Customer Release Agreement from:	Yuji Tsuchiya <yuji_tsuchiya@jp.honda> Takeshi Hara <takeshi_hara@jp.honda>
This product is intended for use in:	AP2 engine, 1.5L, 4Cyl, MY27

The product has been developed, validated, and released exclusively for use in these applications.
The product has not been developed, validated, and released for use outside of the applications specified above. The following areas in particular fall outside the area of released applications.

BOSCH	BOSCH	HONDA
PIC	approved	approved
Sang_HuynhMinh Digitally signed by Sang_HuynhMinh Date: 2026.03.25 15:51:37 +07'00'	Kosuke_Nakazono Digitally signed by Kosuke_Nakazono Date: 2026.03.31 09:10:04 +09'00'	

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1.1 Referenced documents and files

Table 1: Referenced documents

Software	
Description	**File Name**
Release Note SW	62XS-E-S019_MG1CS022_62XSC_V036_TPROT_Release_Note_SW_Series.pdf
Executable file/s (ASCII encoded binary executable code and data)	37820-62X-MD1: 10SW132477_cleaned.hex / 10SW132477.enc 37820-62X-MH1: 10SW132478_cleaned.hex / 10SW132478.enc 37820-62X-ME1: 10SW132479_cleaned.hex / 10SW132479.enc 37820-62X-MF1: 10SW132480_cleaned.hex / 10SW132480.enc 37820-62X-MG1: 10SW132481_cleaned.hex / 10SW132481.enc 37820-62X-MJ1: 10SW132482_cleaned.hex / 10SW132482.enc
A2L file (description of calibration values, signals, communication methods and interfaces)	37820-62X-MD1: 10SW132477.a2l 37820-62X-MH1: 10SW132478.a2l 37820-62X-ME1: 10SW132479.a2l 37820-62X-MF1: 10SW132480.a2l 37820-62X-MG1: 10SW132481.a2l 37820-62X-MJ1: 10SW132482.a2l
Software documentation	MG1CS022_62XSC_V036_TPROT_0_EN_SWDoc.pdf
Delta software documentation (changes with regard to the previous delivery)	Please refer to SRS of items list at “7. Changed issues from last delivery”
Target resources consumption	MG1CS022_62XSC_V036_TPROT_SMART_Report.xlsx
List of implemented requirements (new, changed and corrected functionality)	Please refer to “7. Changed issues from last delivery”
List of new, changed, removed labels	MG1CS022_62XSC_V036_MG1CS022_62XSC_V036_TPROT_difference_summary
Known bugs and bug fix roadmap	None
Open issues	None
Applicable PROF configuration	For XETK ECU: Honda_MG1_XETK_Prof_CS_woOTA_Dev_v2.0.1.zip
Bypass data	None
Customer specific documents & files…	None

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2 Intended use

The product is released based on the regulatory requirements directly applicable to the product at the time of this document creation in the following target market(s):

Brazil/Paraguay

Deliveries and services (fulfillment of contract) shall be subject to the provision that there are no obstacles to performance due to national or international (re-)export control regulation, in particular embargoes or other sanctions.
The product can be used in a released target market in the applications set out chapter 1, subject to the limits, conditions and other specifications described in this chapter ("Intended use").

In the case that the customer wants to use the product outside the Intended Use, the customer shall

(1) on its own responsibility evaluate and comply with any regulatory requirements resulting there from for the product and evaluate and ensure the usability of the product for the area of application intended by the customer or

(2) obtain a new, extending Bosch release, which shall be ordered separately from Bosch (e.g. by means of a change request).

The software is solely to be used in connection with the system components, mounting and environment conditions which have been validated and released together with this configuration.

Table 2: Variants

Program Version (SW)	Injection system	Engine	Vehicle
MG1CS022_62XSC / V036_TPROT	GDI	AP2 FFV, 1.5L, 4Cyl	62XS

2.1 Software particularities and restrictions

- After release of a new SW version for series production, all product lines at HONDA which were jointly planned between Bosch and HONDA for taking over that SW-version shall be replaced by the new SW version as soon as possible (“Time Period”).
  In case of additional variant substructure within HONDA’s product line handling, it is agreed that HONDA within the Time Period shall take over the new SW version also for such additional variant substructures in accordance with the joint planning for the product lines.
  The release of any software versions preceding the new SW version will expire for the affected product lines and potential additional variant substructures after the Time Period has elapsed.

- Software Update reprogramming (Reflash) routines are implemented. These are protected with anti-tampering measures according to requirements agreed with the customer. The customer shall validate the implemented measures to prevent unauthorized reprogramming. If necessary additional mechanism(s) in the product or by an upper security layer, to comply with relevant legal requirements shall be implemented by the customer.

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2.2 Software remarks

- The EEPROM layout is compatible to the release predecessor program version (development baseline).

2.3 Software compliance information

The software documentation includes compliance information:

- General compliance information
- “Algorithms to Review”-chapters showing general compliance information, including Compliance-related calibration hints.
- Security and Data Protection Information

The compliance information includes information, in regard to the utilization of the ECU and the software in the OEM’s vehicle. Any compliance information shall not dispend the OEM from its responsibility and its obligation (i) to verify and demonstrate the conformity of the application of the calibrated software in the vehicle with any applicable law and (ii) to comply with relevant type approval and disclosure requirements under any applicable law.

The compliance information is based on and limited to our current actual knowledge. It is based on the assumption that content of the software which cannot be viewed and assessed by Bosch is designed, implemented and calibrated in compliance with applicable law and does not use software functions of this release improperly in other software functions of the ECU or other control units.

Below are the changes of the compliance information in this SW version compared to the predecessor SW version.

General compliance information (see chapter I in the Technical SW documentation - MG1CS022_62XSC_V036_TPROT_0_EN_SWDoc.pdf)

“Algorithms to review” chapters (see chapter IV in the technical SW documentation - MG1CS022_62XSC_V036_TPROT_0_EN_SWDoc.pdf)

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IV Algorithms to Review

1 Overview List of Algorithms to Review

[AEVAB 17.10.0;0/aevab] Algorithm to review : [Algorithm_to_review] 12068
[AirDev 1.0.0;0 42.90.0_1.2.0;0/airdev_cmpnprotn 1.0.0;0] Algorithm to review : Actuator component protection [airdev_cmpnprotn] 105
[AirDvP 666.22.0_1.3.0;0/airdvp 666.20.0;2] Algorithm to review: Switching to substitute values [codex_software_compliance_airdvp_fid] 476
[AirMod 673.3.0_1.1.0;0/airmod_scav 666.12.0;1] Algorithm to review : Scavenging [airmod_scav] . 596
[ASDCtl 76.8.0;1/asdctl_reffildstbcctrl 1.0.0;1] Algorithm to review : [] 929
[BGKV 18.19.0;2/bgkv 18.14.0;1] Algorithm to review : On Board Fuel Consumption Monitoring [alg2rv_bgkv] 11964
[BGPSMAX 14.1.0;0/bgpsmax] Algorithm to review : Setpoint maximum intake manifold pressure [BGPSMAX.MAIN.LEAN_MODES.] 2012
[ChrCtl 666.9.0;1 678.2.0;0/chrctl_chrgprdn 666.10.0;1] Algorithm to review : [chrctl_chrgprdn] ... 1800
[ChrSet 674.4.0;0/chrset_chrglimn 666.4.0;1] Algorithm to review : [chrset_chrglimn] 1976
[ChrSet 674.4.0;0/chrset_egrdeactvn 666.4.0;3] Algorithm to review : [chrset_egrdeactvn] 1977
[ChrSet 674.4.0;0/chrset_egrsp 666.4.0;3] Algorithm to review : [chrset_egrsp] 1977
[ChrSet 674.4.0;0/chrset_scav 666.4.0;2] Algorithm to review : [chrset_scav] 1975
[ChrSet 674.4.0;0/chrset_soundintkvlv 666.4.0;0] Algorithm to review : [chrset_soundintkvlv] 1976
[CoEng 8.35.0_4.1.0;0/coeng_ctrleom 1.0.0;1] Algorithm to review : [] 2046
[CoEng 8.35.0_4.1.0;0/coeng_detopteom 1.0.0;0] Algorithm to review : [] 2045
[CoEng 8.35.0_4.1.0;0/coeng_fco 1.0.0;1] Algorithm to review : [] 2046
[CoEng 8.35.0_4.1.0;0/coeng_strtstop 1.0.0;2] Algorithm to review : [] 2047
[DASwCpt_ActrOutpCnvn 2.0.0;0/daswcpt_actroutpcnvn] Algorithm to review : [alg_to_rev] 4376
[DASwCpt_ActrPwrStgDrvr 2.0.0;0/daswcpt_actrpwrstgdrvr] Algorithm to review : [alg_to_rev] 4378
[DASwCpt_AdcInp 2.1.0;0/daswcpt_adcinp] Algorithm to review : [alg_to_rev] 4381
[DASwCpt_PwmInp 2.1.0;0/daswcpt_pwminp] Algorithm to review : [alg_to_rev] 4388
[DASwCpt_SentAcqn 2.0.0;0/daswcpt_sentacqn] Algorithm to review : [alg_to_rev] 4394
[DASwCpt_SigIvt 1.0.0;1/daswcpt_sigivt] Algorithm to review : [alg_to_rev] 4360
[DASwCpt_SpArbn 1.0.0;0/daswcpt_sparbn] Algorithm to review : [alg_to_rev] 4361
[DELib 1.67.0_1.7.0;0/delib_actrcptprotn 1.1.0;0] Algorithm to review : Actuator Component Protection [Actuator_Component_Protection_FS] 4359
[DELib 1.67.0_1.7.0;0/delib_limndiagcactrsnsr 1.1.0;0] Algorithm to review : Limitation of diagnostic functions for actuators and sensors [Limitation_of_diagnostic_functions_for_actuators_and_sensors] 4359
[DELib_GlbArAdpr 2.0.0;0/delib_glbaradpr] Algorithm to review : [alg2rev] 4816
[DELib_GovrDeMonLib 4.4.0;0/delib_govrdemonlib] Algorithm to review : Deactivation of the governor deviation monitoring to avoid false diagnosis [DisableGovDvtMon] 5107
[DELib_GovrDeMonLib 4.4.0;0/delib_govrdemonlib] Algorithm to review : Governor Deviation Monitoring [Governor_Deviation_Monitoring] 5107
[DSCHED_Cal 30000.2.0;0/dsched_cal 1.2.0;1] Algorithm to review : Diagnostic Scheduler, Priority Computation [ ] 7514
[DSCHED_Eval 30000.13.0;1/dsched_eval 30000.11.0;1] Algorithm to review : Diagnostic Scheduler, Priority Computation [ ] 7529
[EGRVlv_DELib_GovrDeMonLib 4.4.0;0/egrvlv_delib_govrdemonlib_auto_fs] Algorithm to review : Deactivation of the governor deviation monitoring to avoid false diagnosis [DisableGovDvtMon] ... 4735
[EGRVlv_DELib_GovrDeMonLib 4.4.0;0/egrvlv_delib_govrdemonlib_auto_fs] Algorithm to review : Governor Deviation Monitoring [Governor_Deviation_Monitoring] 4735
[EGSDCmn_CoorrParlShoTrip 1.8.1;3/egsdcmn_coorrparlshotrip] Coordinator of short trip requests for parallelization of active exhaust gas sensor and catalyst diagnosis 7931
[EGSDia 1.0.0;5 1.68.0;1/egsdia_actvdiag 1.0.0;5] Algorithm to review : Active diagnoses of exhaust gas sensor dynamic and catalyst [egsdia_actvdiag] 7771
[EGSDia 1.0.0;5 1.68.0;1/egsdia_detnlamoffssens 1.1.0;1] Algorithm to review : Calculation of a Lambda offset of a wideband Lambda sensor for adaptation in Lambda control [egsdia_detnlamoffssens] 7772
[EGSPrepn 1.0.0;4 1.26.0;0/egsprepn 1.0.0;4] Algorithm to review : Tolerance reduction of lambda signal [] 8254
[EngStrt 100.20.0_1.14.0;0/engstrt_engstopctrl 1.0.0;2] Algorithm to review : [engstrt_engstopctrl] 8645
[EngStrt 100.20.0_1.14.0;0/engstrt_engstrt 1.0.0;1] Algorithm to review : [engstrt_engstrt] 8645
[EngStrt 100.20.0_1.14.0;0/engstrt_engwrmup 1.0.0;2] Algorithm to review : [engstrt_engwrmup] . 8646
[EngStrt 100.20.0_1.14.0;0/engstrt_fuqlyadpn 1.1.0;0] Algorithm to review : [engstrt_fuqlyadpn] .. 8646

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[EngStrt 100.20.0_1.14.0;0/engstrt_fuwtrevapoil 1.0.0;3] Algorithm to review : [engstrt_fuwtrevapoil] 8647
[EngStrt 100.20.0_1.14.0;0/engstrt_noise 1.1.0;1] Algorithm to review : [engstrt_noise] 8647
[EngStrt 100.20.0_1.14.0;0/engstrt_strtstop 1.0.0;4] Algorithm to review : [engstrt_strtstop] 8648
[EngStrt_StrtSysRels 1.21.0;2/engstrt_strtsysrels] Algorithm to review : [Table] 8782
[ExhCo 0.6.0;1/exhco_main_gs 0.6.0;0] Algorithm to review :Exhaust co–ordinator(Gasoline) 9936
[ExhMgT 1.0.0;2 119.25.0_2.1.0;0/exhmgt_catheatg 1.0.0;3] Algorithm to review : Unified Main-Alg2Rv for DS+GS [CatHeatg] 9945
[ExhMgT 1.0.0;2 119.25.0_2.1.0;0/exhmgt_cmbenglimn 1.0.0;0] Algorithm to review : Combustion Engine Limitation [CmbEngLimn] 9944
[ExhMgT 1.0.0;2 119.25.0_2.1.0;0/exhmgt_exhtcptprotn_gs 1.0.0;0] Algorithm to review : Exhaust Temperature Component Protection (GS) [ExhMgT_ExhTCptProtn_GS] 9945
[ExhMgT 1.0.0;2 119.25.0_2.1.0;0/exhmgt_scavexhtmngt_gs 1.0.0;2] Algorithm to review : Scavenging in exhaust temperature management (GS) [ScavExhTMngt_GS] 9943
[FlexFl 3.1.0;0 103.1.0;0/coldstrtprm_coor 103.6.0;1] Algorithm to review : Market specific request to temporary disable diagnostics in event of fuel change detected or special conditions [diagnostic_disable_obdbr2] 10834
[FlexFl 3.1.0;0 103.1.0;0/ffprm_ethcalc 3.1.0;0] Algorithm to review : Market specific request to temporary disable diagnostics in event of fuel change detected or special conditions [diagnostic_disable_obdbr2] 10834
[FIMA 33.15.0_1.4.0;0/fumixadpn 1.0.0;1] Algorithm to review : [FuMixAdpnAlgToRvTbl] 10912
[FIMA 33.15.0_1.4.0;0/fumixadpndiag 1.0.0;1] Algorithm to review : [FuMixAdpnDiagAlgToRvTbl] .. 10912
[FlSplG 65.15.0_5.9.0;0/flsplg_dkvbde 1.0.0;0] Algorithm to review : [] 11031
[FlSplG 65.15.0_5.9.0;0/flsplg_hdpcontrol 1.0.0;0] Algorithm to review : [] 11032
[FlSplG 65.15.0_5.9.0;0/flsplg_hdrpsol_hdrpsoladp 1.0.0;0] Algorithm to review : [] 11031
[FlSplG 65.15.0_5.9.0;0/flsplg_pressuredetection 1.0.0;0] Algorithm to review : [] 11032
[GEVCtl 666.5.0;2 46.0.0_3.3.0;0/gevctl_scav 666.5.0;2] Algorithm to review : [gevctl_scav] 11214
[GEVCtl 666.5.0;2 46.0.0_3.3.0;0/gevctl_vvtcountermeasure 666.5.0;0] Algorithm to review : [gevctl_vvtcountermeasure] 11215
[GEVCtl 666.5.0;2 46.0.0_3.3.0;0/gevctl_vvtenagcdn 666.5.0;1] Algorithm to review : [gevctl_vvtenagcdn] 11214
[GEVCtl 666.5.0;2 46.0.0_3.3.0;0/gevctl_vvtobdadpn 666.5.0;1] Algorithm to review : [gevctl_vvtobdadpn] 11214
[GEVCtl 666.5.0;2 46.0.0_3.3.0;0/gevctl_vvtsp 666.5.0;2] Algorithm to review : [gevctl_vvtsp] 11213
[HEGO 1.71.1_1.0.0;0/hego_main_gs 1.0.0;1] Algorithm to review : Binary oxygen sensor heater control [hego_main_gs] 11510
[I15031_srv9 30000.30.0;2/i15031_srv9 30000.27.0;1] Algorithm to review : [] 5243
[IgCtl 91.4.0;1/igctl_ignagspdetrm 1.0.0;0] Algorithm to review : ignition angle set point determination [] 11664
[IKCtl 24.18.0;2/preigncptprotn 24.0.0;1] Algorithm to review : Pre–Ignition Component Protection [] 11772
[InjSyG 159.1.0_1.1.0;0/injsyg_fco 1.0.0;1] Algorithm to review : Fuel Cut–Off [InjSyG_Fco_Main-Alg2Rv] 11961
[InjSyG 159.1.0_1.1.0;0/injsyg_fuinjdurn 1.0.0;2] Algorithm to review : Fuel Injection Duration [InjSyG_FuInjDurn_MainAlg2Rv] 11962
[InjSyG 159.1.0_1.1.0;0/injsyg_fuinjtmngsp 1.0.0;2] Algorithm to review : Fuel injection timing setpoint [InjSyG_FuInjTmngSp_MainAlg2Rv] 11961
[InjSyG 159.1.0_1.1.0;0/injsyg_noise 1.0.0;1] Algorithm to review : Generation of brand specific sound [InjSyG_Noise_MainAlg2Rv] 11962
[KnDetControl 18.13.0;2/kndetcontrol] Algorithm to review : 13645
[LamCL 9.16.0_1.6.0;0/lamcl_enlnmtprotn 1.1.0;0] Algorithm to review : [Enleanment Protection] . 13665
[LamCL 9.16.0_1.6.0;0/lamcl_lamctrl 1.2.0;2] Algorithm to review : [Lambda Control] 13665
[LamCL 9.16.0_1.6.0;0/lamcl_lamsp 1.1.0;0] Algorithm to review : [Lambda Setpoint] 13665
[LamCL 9.16.0_1.6.0;0/lamcl_scav 1.1.0;0] Algorithm to review : [Scavenging] 13664
[LRS 1.58.1_1.0.1;0/lrs] Algorithm to review : Continuous lambda control [Continuous_lambda_control] 13739
[PwrTrn 287.6.0;0/ptcoorndyn_tqlead 135.7.0;2] Algorithm to review : [PtCoornDyn_TqLead1] 15800
[SpdCtl 1.2.0;1 206.4.0;0/spdctl_cntrlengspdsp 1.2.0;1] Algorithm to review : [SpdCtl_CntrlEngSpdSp] 16519
[SpdCtl 1.2.0;1 206.4.0;0/spdctl_engspdmaxlimn 1.0.0;0] Algorithm to review : Engine Speed Maximum Limitation [spdctl_engspdmaxlimn] 16520
[SpdCtl 1.2.0;1 206.4.0;0/spdctl_tqresv 1.2.0;0] Algorithm to review : [spdctl_tqresv] 16520
[SysDiag_TestCoorr 4.0.0;4/sysdiag_testcoorr] Algorithm to review : FC: SysDiag_TestCoorr(special user sequence) 17934

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[TCE 123.17.0_1.8.0;0/tce_fuinjccodurgovrrn 123.10.0;2] Algorithm to review : [tce_fuinjccodurgovrrn] 17944
[TME 159.1.0_1.1.0;0/tme_fuinjccodurgovrrn 1.0.0;0] Algorithm to review : Fuel injection cut-off during overrun [tme_fuinjccodurgovrrn] 18750
[TME 159.1.0_1.1.0;0/tme_trftqreqtocmbsp 1.0.0;0] Algorithm to review : Transfer requested torque to combustion set points [tme_trftqreqtocmbsp] 18750
[TWC 1.69.0;1/twc_actvdiag 1.0.0;0] Beobachtungswürdige Algorithmen : Active diagnosis of Three-Way-Catalyst (TWC) [twc_actvdiag] 18897
[TWC 1.69.0;1/twc_lamctrl 1.0.0;1] Beobachtungswürdige Algorithmen : Air/Fuel ratio (lambda) control of Three-Way-Catalyst (TWC) [twc_lamctrl] 18898
[TWCC_CatPurge 1.54.0;8/twcc_catpurge] Catalyst Purge 18899
[UEGO_OperRelsWoDewPnt 1.15.0;1/uego_operrelswodewpnt 1.10.0;0] Algorithm to review : Sensor operation before engine start [Algorithm_to_review_Sensor_operation_before_engine_start] ... 20634
[ZWMINBG 40.4.0;6/zwminbg] Algorithm to review : 11721

3 Functional safety

ISO26262 Software – Conformance status:

Software with ASIL rating

Bosch points out that the ASIL-classified requirements as per ISO 26262, their implementation and the assumptions made for this purpose are documented in the following documents: 62XS-E-S001_Honda_AP2_MY27_City_MG1_DIA_signed.pdf

The customer has to ensure that the software is suitable to be used within the overall system regarding any functional safety requirements assigned to the software in the overall system.

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4 Data protection and cybersecurity aspects

4.1 General information about cybersecurity

The product was developed within an ISO/SAE 21434 certified Cyber Security Management System and corresponding cybersecurity activities were done.

4.2 Cybersecurity claims

The product contains functions which could processes personal data. If so, the product retains the risk of discloser of person identifiable information. As the data controller, the customer designs the processing in detail. It is expected that the customer evaluates if the protective measures of the product are sufficient for the dedicated use case.

Irrespective of the high quality of the delivered component, the product retains the risk that currently unknown cybersecurity vulnerabilities get discovered in the future. As risk mitigation Bosch has installed a Cybersecurity monitoring and further continual cybersecurity activities according to ISO/SAE 21434. In case a cybersecurity vulnerability in a Bosch product is discovered or disclosed by or to the customer it is highly recommended to inform Bosch up-front its publication via PSIRT@bosch.com.

The product retains the risk that intellectual property (“IP”) from Bosch, the customer and in some cases 3rd parties can be disclosed during the software distribution. For the purpose of IP protection and avoidance of re-engineering, it is expected that delivered software is handled as confidential during the distribution process to the customer, into plants, workshops. It is expected that the customer has implemented appropriate measures within his Customer’s Software Logistic Infrastructure (IT network) and associated processes to secure and restrict the access to delivered software.

The product retains the risk of discloser of confidential data (including but not limited to personal data) during the decommissioning process of the customer or 3rd parties. It is expected that the customer evaluates and adequately mitigates the risk on vehicle level or within the logistic infrastructure.

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5 Safety and warning notes

This paragraph contains important safety and warning notes for handling and operating the ECU in series applications.

Default reactions and fault clear: Default reactions caused by monitoring and specific inhibit-reactions in the OBD system may be active even after fault clear for the remaining power on cycle. Having these default reactions in the control unit active without any entry in the fault memory being accessible by a Generic OBD Scan Tool has to be avoided because this could result in an OBD-non-compliance. For that reason, the notice shall be given that after a fault clear by a service tester or the Generic OBD Scan Tool, a new power-up cycle shall be started.

The connection of the ECU to vehicles or engines, which do not correspond to the validated configuration, may lead to damage or malfunction of the or connected components or non-conformance to the legal requirements, or to hazards. E.g. this may occur by mistake, if different types of control units with identical electrical connectors are used.

The mounting position and the installation of the ECU in the vehicle should be selected such that the conditions described in this TCD and the associated documents are adhered to. An inappropriate installation can lead to damage of the control unit (and to malfunctions resulting from it). Specifically, damage could result from inadmissible temperature, vibration loads or the effect of specific substances. Additional instructions for the installation of the control unit are given in chapter 4.2.3 from Referenz TCD HW.

ECU designed for use with main relay are only allowed to be used in combination with main relay (reverse polarity protection, quiescent current). Without main relay and in case of reverse polarity the connected actuators may be supplied with negative current. This may lead to damage of actuators or other hazards.

The used actuators are assumed being intrinsically safe against wrong control (e.g. actuation time too long). As an alternative heat shield are used that prevent fire from spreading to the vehicle.
Fire protection: Wrong control of sensors/actuators implies danger of fire by their electrical overload (1.) as well as by the effect of the sensor/actor (2.).
1. Wrong control of actuators causing overheating of external components:
Wrong control of actuators (e.g. too high control values or too long activation) may lead to overheat of external components.
Other technologies/external measures: Overheating shall be prevented by sufficient design of the external components or supply lines (including e.g. heat shields).

2. Wrong control of actuators causing overheat of exhaust system:
Engine combustion is determined by the control of several actuators (e.g. to provide a specific amount of fresh air into the cylinder). Wrong control of theses actuators may lead to wrong engine operation and cause overheat of the exhaust system. Endangered are in particular adjacent electrical, hydraulic or fuel lines. There may also be a fire hazard for highly inflammable materials below the vehicle (e.g. dry grass).
Other technologies/external measures: Protection of the vehicle (including e.g. heat shield for 3 way catalyst (gasoline), to mitigate the effect of a high engine / exhaust system temperature to prevent fire starting in the vehicle.

A short circuit in either wiring harness or other components may lead to high current in the supply line which may cause a fire hazard. The max. output currents for which the intact ECU limits the current, are specified in this TCD. The wiring harness must be designed for this current level. If applicable, the I/O description also designates pins without current limitation. The OEM shall ensure appropriate dimensioning of wiring diameters and fuses to protect the vehicle against fire hazard.

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Connecting sensor ground lines elsewhere as specified (e.g. engine, vehicle body) may lead to malfunction, diagnostic fault codes or damage caused by ground offsets.
Sensor grounds must be connected to the dedicated sensor ground pins provided by the ECU.

A control unit with integrated fan control may activate the fan while the engine hood is open, even if the engine is not running. Please consider this potential risk in the vehicle's service manual and consider the placement of a respective warning label in the engine compartment.

Isolated housing of Solenoid ECU.
In case of an ECU HW defect which leads to a conducting connection of the booster-C and the ECU housing the electrical potential of this housing will be increased to Booster-C potential (up to 260V). Touching the ECU housing with this defect is dangerous due to high voltage. That is why the housing of the ECU is always connected to chassis ground potential, internally via the ECU connector or externally by other means. Consequently, a robust connection to chassis ground potential in the harness has to be ensured.

Pre-drive activation of the electrical fuel pump
In applications with ECU pre-drive activation, e.g. by wake up or remote control, the electric fuel pump may be activated prior to "ignition on" and therefore fuel pressure may build up even with the engine still off. Appropriate service instructions are required, specifically with regard to hazards of injury or fire in case of servicing a potentially pressurized fuel system.

- High voltage at injection power stages and ignition output system (gasoline). There is a risk of electrical shock, by touching active live parts, ECU output pins, wiring harness connector, injection valves.
- Gasoline solenoid system typically >60V
- Ignition for gasoline system, output of the ECU typically > 400V up to spark plug typically > 10.000V

A possible measure is e.g. to use a warning sign (label) in the engine compartment. During service work the system must be deactivated (T15 off). If some work has to be done with Ignition ON, the service manual has to inform about the handling and safety procedure.

During operation of system, high magnetic and electrical near-fields of common rail systems (injection power stage) and ignition output system can appear. This may present a risk for people with cardiac pacemakers and other electronic implants.
- Near-field of gasoline solenoid power stages (pulse voltage with high currents)
- Near-field of Ignition systems for gasoline systems. (pulse voltage with high currents)

The wiring harness design shall ensure appropriate protection levels. Compliance with limits for electric and magnetic near fields has to be tested specifically for each vehicle or wiring harness variant. It is recommended to provide a respective warning label in the engine compartment. During service the system must be deactivated (terminal 15 off). If work has to be done with Ignition on or the engine running, the service manual shall inform about appropriate safety procedures.

A wrongly full open throttle valve may lead to unmanageable increase in torque. In case of mechanical throttle systems the maximum air mass via the bypass must be mechanically limited so, that the maximum additional torque by bypass is not safety relevant, and thus even in the case of an erroneous control of the bypass valve, no risk of functional safety can occur.

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6 Appendices and references

List of abbreviations and terms

Table 3: List of abbreviations

Abbreviation	Meaning
ECU	Electronic Control Unit / Engine Control Unit
EEPROM	Electrically erasable programmable read-only memory, is a type of non-volatile memory which can be erased and reprogrammed repeatedly by electrical voltage.
HW	Hardware
IP	intellectual property;
OEM	Original equipment manufacturer
SW	Software
TCD	Technical customer documentation
T15	Terminal 15; terminal for switching on of Control Unit

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7 Changed issues from last delivery

RQONE04989768	[62XS SRL#1674] Series PVER configuration update for TPROT
SRS: None RAG: None	GConf / PCFG_MG1CS022_62XSC_33

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8 History

Table 4: Change/History

Release	Date	Edited by	Release description
1.0	2026/03/25	Bosch/Huynh Minh Sang	For 62XS Series container release.

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9 Contact person

- Contact person: MS/EJV4-PS, Huynh Minh Sang

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